

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250272381

Kind Code

A1

Publication Date

August 28, 2025

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SECURITY DEVICE AND SYSTEM-ON-CHIP INCLUDING SECURITY DEVICE

Abstract

An example security device for controlling access authority of a user device includes a mode controller, an authentication controller, and an access controller. The mode controller is configured to set, based on a one-time programmable (OTP) value, a security mode indicating a method of authenticating the access authority of the user device. The authentication controller is configured to determine, based on the security mode, whether to allow access of the user device. The access controller is configured to activate or deactivate the access of the user device based on a result of the determination of the authentication controller.

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Family ID: 1000008340112

Appl. No.: 18/972430

Filed: December 06, 2024

Foreign Application Priority Data

KR

10-2024-0027507

Feb. 26, 2024

Publication Classification

Int. Cl.: G06F21/44 (20130101); G06F21/60 (20130101); G06F21/74 (20130101)

U.S. Cl.:

CPC G06F21/44 (20130101); G06F21/604 (20130101); G06F21/74 (20130101);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2024-0027507, filed on Feb. 26, 2024, in the Korean Intellectual Property Office, the disclosure of which is incorporated by reference herein in its entirety.

BACKGROUND

[0002] Recently, security attacks on a device including a plurality of intellectual properties (IPs), such as a system-on-chip (SoC), have increased significantly. Accordingly, security requirements for an SoC have also increased. The SoC may include a security device for determining whether to allow the access of a user device used by a user performing debugging on a plurality of IPs.

[0003] The security device may use a password, a digital signature, or the like to determine whether to allow the access of the user device. A method using a password may provide higher user convenience but lower security strength than a method using a digital signature. Conversely, the method using a digital signature may provide higher security strength but lower user convenience than the method using a password. Thus, a suitable method for the security device to determine whether to allow the access of the user device is desired.

SUMMARY

[0004] The present disclosure relates to a security device that uses an authentication method suitable for situations.

[0005] In some implementations, a security device for controlling access authority of a user device includes a mode controller configured to set, based on a one-time programmable (OTP) value, a security mode indicating a method of authenticating the access authority of the user device, an authentication controller configured to determine, based on the security mode, whether to allow access of the user device, and an access controller configured to activate or deactivate the access of the user device based on a result of the determination of the authentication controller.

[0006] In some implementations, a security device for controlling access authority of a user device to a plurality of intellectual properties (IPs) includes a mode controller configured to set a security mode of the user device for each of the plurality of IPs based on each of a plurality of OTP values, an authentication controller configured to determine, based on the security mode, whether to allow access of the user device to each of the plurality of IPs, and an access controller configured to activate or deactivate the access of the user device to each of the plurality of IPs based on a result of the determination of the authentication controller.

[0007] In some implementations, a system-on-chip (SoC) includes a plurality of IPs and a security device configured to control access authority of a user device to the plurality of IPs, wherein the security device includes a mode controller configured to set, based on an OTP value, a security mode indicating a method of authenticating the access authority of the user device to the plurality of IPs, an authentication controller configured to determine, based on the security mode, whether to allow access of the user device to the plurality of IPs, and an access controller configured to activate or deactivate the access of the user device to the plurality of IPs based on a result of the determination of the authentication controller.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] Implementations will be more clearly understood from the following detailed description taken in conjunction with the accompanying drawings.

[0009] FIG. 1 is a block diagram illustrating an example of a security system.

[0010] FIG. 2 is a block diagram illustrating an example of a system-on-chip (SoC) including a security device.

[0011] FIG. 3 is a diagram illustrating an example of a security mode corresponding to a one-time programmable (OTP) value of a security device.

[0012] FIG. 4 is a block diagram illustrating another example of an SoC including a security device.

[0013] FIG. 5 is a diagram illustrating an example of an OTP value and a security mode corresponding to a plurality of intellectual properties (IPs) connected to a security device.

[0014] FIG. 6 is a flowchart illustrating an example of an operating method of a security device.

[0015] FIG. 7 is an example flowchart illustrating the operation of a user device and a security device when the security mode of the security device is a first security mode.

[0016] FIG. 8 is an example flowchart illustrating the operation of a user device and a security device when the security mode of the security device is a second security mode.

[0017] FIG. 9 is an example flowchart illustrating the operation of an authentication server, a user device, and a security device when the security mode of the security device is a third security mode.

[0018] FIG. 10 is an example flowchart illustrating the operation of an authentication server, a user device, and a security device when the security mode of the security device is a fourth security mode.

DETAILED DESCRIPTION

[0019] Hereinafter, Implementations will be described in detail with reference to the accompanying drawings.

[0020] FIG. 1 is a block diagram illustrating an example of a security system.

[0021] Referring to FIG. 1, a security system may include a user device **10**, an authentication server **20**, and a system-on-chip (SoC) **30**.

[0022] The security system may be a system including components necessary for a user to perform debugging on a plurality of intellectual properties (IPs) **200** included in the SoC **30**.

[0023] The user device **10** may include hardware, software, or firmware for debugging the plurality of IPs **200** included in the SoC **30**. The user may input data necessary to authenticate the access authority of the user to the plurality of IPs **200** included in the SoC **30**, through the user device **10** to the security device **100** included in the SoC **30**. For example, the user may input a password through the user device **10** to the security device **100** included in the SoC **30**.

[0024] The authentication server **20** may be a server for generating a digital signature used to authenticate the access authority of the user device **10** to the plurality of IPs **200** included in the SoC **30**. For example, when receiving an identification (ID) from the user device **10**, the authentication server **20** may identify the authority of a user corresponding to the ID and generate a digital signature corresponding to the user.

[0025] The SoC **30** may include the security device **100** and the plurality of IPs **200**.

[0026] The security device **100** may provide a function of driving all external pins of the plurality of IPs **200** included in the SoC **30** or reading values therefrom. The security device **100** may provide an interface for performing debugging on the plurality of IPs **200**, to the user device **10** used by the user whose access authority have been authenticated.

[0027] In some implementations, the security device **100** may include a one-time programmable (OTP) memory **111** for storing an OTP value. The OTP value may be a value used to set a security mode indicating a method of authenticating the access authority of the user device **10**. The security device **100** may set a security mode based on the OTP value and determine whether to allow the access of the user device **10**, based on the security mode. More detailed operations of the security device **100** will be described below with reference to FIG. 2.

[0028] The plurality of IPs **200** may include a device for processing at least one operation necessary to perform various operations of the SoC **30** and may include, for example, an

application processor (AP). The plurality of IPs **200** may be debugged through the user device **10**.
[0029] FIG. **2** is a block diagram illustrating an example of an SoC including a security device.
[0030] Referring to FIG. **2**, an SoC **30** according to some implementations may include a security device **100** and first to p th IPs **200_1** to **200_p** (where “ p ” is a natural number of 2 or more).
[0031] The security device **100** may include a mode controller **110**, an authentication controller **120**, and an access controller **130**.

[0032] In some implementations, the mode controller **110** may set a security mode indicating a method of authenticating the access authority of the user device **10** based on an OTP value.
[0033] The OTP value may be a value indicating which security mode is a security mode indicating a method of authenticating the access authority of the user device **10** by the authentication controller **120**. The OTP value may be stored in an OTP memory **111** included in the mode controller **110**.

[0034] In some implementations, a value of each of the bits included in the OTP value may be changeable in one direction. Here, the one direction may refer to any one of a direction in which a value of each of the bits included in the OTP value increases and a direction in which the value decreases. Each of the bits included in the OTP value may have any one of a first logic value (e.g., logic 1) and a second logic value (e.g., logic 0). In this case, the bits included in the OTP value may all be initially set to have the second logic value and may be changed from the second logic value to have the first logic value but may not be changed from the first logic value to have the second logic value.

[0035] For example, the OTP memory **111** may store the values of the bits included in the OTP value through a plurality of fuses. When physically connected, each of the plurality of fuses may indicate the second logic value. On the contrary, when physically disconnected, each of the plurality of fuses may indicate the first logic value. In this case, the plurality of fuses may not be reconnected after being physically disconnected. Thus, each of the plurality of fuses may be changed from the state of indicating the second logic value to the state of indicating the first logic value but may not be changed from the state of indicating the first logic value to the state of indicating the second logic value.

[0036] In some implementations, the OTP value may include $(n-1)$ bits (where “ n ” is a natural number of 2 or more). In this case, the mode controller **110** may set the security mode to any one of first to n th security modes based on the number of bits with the first logic value among the bits included in the OTP value. For example, when the number of bits with the first logic value among the bits included in the OTP value is 0, the mode controller **110** may set the security mode to the first security mode. Also, when the number of bits with the first logic value among the bits included in the OTP value is 1, the mode controller **110** may set the security mode to the second security mode. Also, when the number of bits with the first logic value among the bits included in the OTP value is k (where “ k ” is a natural number of 0 to $(n-1)$), the mode controller **110** may set the security mode to a $(k+1)$ th security mode.

[0037] When the OTP value is changed, the mode controller **110** may change the security mode based on the changed OTP value. In this case, the bits included in the OTP value may be changed from the second logic value to the first logic value but may not be changed from the first logic value to the second logic value. Accordingly, the number of bits with the first logic value among the bits included in the OTP value may increase but may not decrease. Thus, the mode controller **110** may change the security mode in a direction in which the index of the security mode (“ k ” in the case of a k th security mode) increases but may not change the security mode in a direction in which the index of the security mode decreases.

[0038] In this case, an i th security mode (where “ i ” is a natural number of 2 to n) may have higher security strength than a j th security mode (where “ j ” is a natural number of 1 to $(n-1)$ and is less than “ i ”). Here, the high security strength may mean that a procedure required to authenticate the access authority of the user device **10** is complex. That is, the mode controller **110** may change the

security mode in a direction in which the procedure required to authenticating the access authority becomes more complex.

[0039] The user may change the security mode of the security device **100** by changing the OTP value stored in the OTP memory **111** by physically disconnecting a plurality of fuses included in the OTP memory **111**. Because the value of each of the bits included in the OTP value may be changed in one direction, the user may change the security mode to increase the security strength of the security device **100**. In this case, by performing debugging on the first to pth IPs **200_1** to **200_p** included in the SoC **30** and then changing the OTP value stored in the OTP memory **111**, the user may change the security mode in a direction in which the security strength of the security device **100** increases.

[0040] Accordingly, the user may perform debugging in a security mode with relatively low security strength before performing debugging on the first to pth IPs **200_1** to **200_p**, thus allowing more convenient authentication, and may set the security mode to have higher security strength by changing the OTP value after performing debugging on the first to pth IPs **200_1** to **200_p**, thereby allowing the debugged details of the first to pth IPs **200_1** to **200_p** to be more securely protected.

[0041] In some implementations, the security mode may include first to fourth security modes, and the mode controller **110** may set the security mode to any one of the first to fourth security modes according to the OTP value. When the security mode includes the first to fourth security modes, an example of the security mode depending on the OTP value and an operation method of the security mode may be as illustrated in FIG. **3**.

[0042] FIG. **3** is a diagram illustrating an example of a security mode corresponding to an OTP value of a security device.

[0043] Referring to FIG. **3**, a table illustrating a security mode depending on the OTP value and an operation method of the security mode may be identified. In the following description with reference to FIG. **3**, the first logic value may refer to logic 1 and the second logic value may refer to logic 0; however, the present disclosure is not limited thereto.

[0044] First, when the OTP value includes 0 (no) bit with the first logic value like “000”, the mode controller **110** may set the security mode to the first security mode. In this case, the first security mode may also be referred to as a bypass mode and may be a mode in which the security device **100** allows the access of the user device **10** without separate authentication. That is, when the security mode is the first security mode, the user may perform debugging on the plurality of IPs **200** included in the SoC **30** even without performing separate authentication.

[0045] When the OTP value includes 1 bit with the first logic value like “001”, the mode controller **110** may set the security mode to the second security mode. In this case, the second security mode may be a mode in which the security device **100** determines whether to allow the access of the user device **10**, through authentication using a password. That is, when the security mode is the second security mode, the user may perform debugging on the plurality of IPs **200** included in the SoC **30** by performing authentication using a password.

[0046] When the OTP value includes 2 bits with the first logic value like “011”, the mode controller **110** may set the security mode to the third security mode. In this case, the third security mode may be a mode in which the security device **100** determines whether to allow the access of the user device **10**, through authentication using a digital signature. That is, when the security mode is the third security mode, the user may perform debugging on the plurality of IPs **200** included in the SoC **30** by performing authentication using a digital signature generated through the authentication server **20**.

[0047] When the OTP value includes 3 bits with the first logic value like “111”, the mode controller **110** may set the security mode to the fourth security mode. In this case, the fourth security mode may be a mode in which the security device **100** determines whether to allow the access of the user device **10**, through authentication using a digital signature and a random value. That is, when the security mode is the fourth security mode, the user may perform debugging on

the plurality of IPs **200** included in the SoC **30** by performing authentication using a random value generated by the security device **100** and a digital signature generated through the authentication server **20**.

[0048] In this case, as illustrated by the arrows on the left side of the table in FIG. 3, the value of each of the bits included in the OTP value may be changed in one direction (from the first logic value to the second logic value).

[0049] Hereinafter, the description will focus on some implementations in which the security mode includes the first to fourth security modes as illustrated in FIG. 3; however, the present disclosure is not limited thereto.

[0050] Referring back to FIG. 2, the authentication controller **120** may determine whether to allow the access of the user device **10**, based on the security mode set by the mode controller **110**.

[0051] The authentication controller **120** may include an encryption circuit **121**, a random number generator **122**, and a memory **123**.

[0052] The encryption circuit **121** may perform an operation related to determining whether to allow the access of the user device **10**. For example, the encryption circuit **121** may generate a hash value of the password by performing a hash operation on the password received from the user device **10** and determine whether to allow the access of the user device **10**, by determining whether the hash value of the password is equal to a prestored value.

[0053] The random number generator **122** may generate a random value used when the security mode set by the mode controller **110** is the fourth security mode.

[0054] The memory **123** may store software used to operate the encryption circuit **121**, the hash value of the password, the random value generated by the random number generator **122**, and/or the like.

[0055] In some implementations, when the security mode is the second security mode, the authentication controller **120** may determine whether to allow the access of the user device **10**, based on the password. When the security mode is the third security mode, the authentication controller **120** may determine whether to allow the access of the user device **10**, based on the digital signature. When the security mode is the fourth security mode, the authentication controller **120** may determine whether to allow the access of the user device **10**, based on the digital signature set based on the random value. A more detailed method of determining whether to allow the access of the user device **10** through the authentication controller **120** in each security mode will be described below with reference to FIGS. 8 to 10.

[0056] The access controller **130** may activate or deactivate the access of the user device **10** based on the determination result of the authentication controller **120**. When the authentication controller **120** determines to allow the access of the user device **10**, the access controller **130** may activate the access of the user device **10**. On the contrary, when the authentication controller **120** determines not to allow the access of the user device **10**, the access controller **130** may deactivate the access of the user device **10**.

[0057] FIG. 4 is a block diagram illustrating another example of an SoC including a security device.

[0058] Referring to FIG. 4, an SoC **30** according to some implementations may include a security device **100** and first to pth IPs **200_1** to **200_p**. The security device **100** may include a mode controller **110**, an authentication controller **120**, and an access controller **130**. Because the main operations of the SoC **30** of FIG. 4 are similar to those described with reference to FIG. 2, the description will focus on differences therebetween.

[0059] In the implementation of FIG. 4, the mode controller **110** may include first to pth OTP memories **111_1** to **111_p**. Each of the first to pth OTP memories **111_1** to **111_p** may store first to pth OTP values. In this case, the number of first to pth OTP memories **111_1** to **111_p** may be equal to the number of first to pth IPs **200_1** to **200_p**. That is, the number of first to pth OTP values may be equal to the number of first to pth IPs **200_1** to **200_p**.

[0060] Each of the first to pth OTP values may be a value indicating which security mode is a security mode indicating a method of authenticating the access authority of the user device **10** to each of the first to pth IPs **200_1** to **200_p** by the authentication controller **120**. For example, the first OTP value may indicate which security mode is a security mode indicating a method of authenticating the access authority of the user device **10** to the first IP, and an mth OTP value (where “m” is a natural number of 1 to p) may indicate which security mode is a security mode indicating a method of authenticating the access authority of the user device **10** to an mth IP.

[0061] The value of each of the bits included in the first to pth OTP values may be changeable in one direction. Each of the bits included in the first to pth OTP values may have any one of a first logic value (e.g., logic 1) and a second logic value (e.g., logic 0). In this case, the bits included in the first to pth OTP values may all be initially set to have the second logic value and may be changed from the second logic value to have the first logic value but may not be changed from the first logic value to have the second logic value.

[0062] In some implementations, each of the first to pth OTP values may include (n-1) bits (where “n” is a natural number of 2 or more). In this case, the mode controller **110** may set the security mode for the first to pth IPs **200_1** to **200_p** to any one of first to nth security modes, based on the number of bits with the first logic value among the bits included in each of the first to pth OTP values. For example, when the number of bits with the first logic value among the bits included in the mth OTP value is 0, the mode controller **110** may set the security mode for the mth IP to the first security mode. Also, when the number of bits with the first logic value among the bits included in the mth OTP value is 1, the mode controller **110** may set the security mode for the mth IP to the second security mode. Also, when the number of bits with the first logic value among the bits included in the mth OTP value is k (where “k” is a natural number of 0 to (n-1)), the mode controller **110** may set the security mode for the mth IP to a (k+1) th security mode. That is, the mode controller **110** may independently set the security mode for the first to pth IPs **200_1** to **200_p**.

[0063] In some implementations, when any one of the first to pth OTP values is changed, the mode controller **110** may change the security mode based on the changed OTP value among the first to pth OTP values. For example, when the mth OTP value is changed, the mode controller **110** may change the security mode for the mth IP **200_m** corresponding to the mth OTP value.

[0064] In some implementations, the authentication controller **120** may determine whether to allow the access of the user device **10** to each of the first to pth IPs **200_1** to **200_p**, based on the security mode. That is, the authentication controller **120** may independently determine whether to allow the access of the user device **10** to each of the first to pth IPs **200_1** to **200_p**, based on the security mode set for each of the first to pth IPs **200_1** to **200_p**.

[0065] In some implementations, the authentication controller **120** may sequentially determine whether to allow the access of the user device **10** to each of the first to pth IPs **200_1** to **200_p**, based on the security mode of each of the first to pth IPs **200_1** to **200_p**. For example, the authentication controller **120** may determine whether to allow the access of the user device **10** to the first IP **200_1** and then determine whether to allow the access of the user device **10** to the second IP **200_2**. However, the present disclosure is not limited thereto, and when a plurality of cores are included in the authentication controller **120**, the authentication controller **120** may simultaneously determine whether to allow the access of the user device **10** to each of the first to pth IPs **200_1** to **200_p**.

[0066] The access controller **130** may activate or deactivate the access of the user device **10** to each of the first to pth IPs **200_1** to **200_p** based on the determination result of the authentication controller **120**. The access controller **130** may independently determine whether to activate the access to each of the first to pth IPs **200_1** to **200_p**, based on the determination result of the authentication controller **120**.

[0067] FIG. 5 is a diagram illustrating an example of an OTP value and a security mode

corresponding to a plurality of IPs connected to a security device.

[0068] Referring to FIG. 5, a table illustrating an OTP value corresponding to the number of each of the first to pth IPs **200_1** to **200_p** included in the security device **100**, a security mode depending on the OTP value, and access activation/deactivation may be identified. In the following description with reference to FIG. 5, the first logic value may refer to logic 1 and the second logic value may refer to logic 0; however, the present disclosure is not limited thereto.

[0069] First, the first OTP value stored in the first OTP memory **111_1** corresponding to the first IP **200_1** may include 0 (no) bit with the first logic value like “000”, and accordingly, the mode controller **110** may set the security mode for the first IP **200_1** to the first security mode. In this case, because the security mode for the first IP **200_1** is the first security mode, the access of the user device **10** to the first IP **200_1** may be activated without separate authentication.

[0070] The second OTP value stored in the second OTP memory **111_2** corresponding to the second IP **200_2** may include 1 bit with the first logic value like “001”, and accordingly, the mode controller **110** may set the security mode for the second IP **200_2** to the second security mode. In this case, because the security mode for the second IP **200_2** is the second security mode, the security device **100** may determine whether to allow the access of the user device **10**, based on the password. In this case, when it is determined that the access is not allowed, the access of the user device **10** to the second IP **200_2** may be deactivated as illustrated in the table.

[0071] The third OTP value stored in the third OTP memory **111_3** corresponding to the third IP **200_3** may include 3 bits with the first logic value like “111”, and accordingly, the mode controller **110** may set the security mode for the third IP **200_3** to the fourth security mode. In this case, because the security mode for the third IP **200_3** is the fourth security mode, the security device **100** may determine whether to allow the access of the user device **10**, based on the random value and the digital signature. In this case, when it is determined that the access is allowed, the access of the user device **10** to the third IP **200_3** may be activated as illustrated in the table.

[0072] The same method may also be applied to the fourth to pth IPs **200_4** to **200_p**.

[0073] As illustrated in FIG. 5, the security mode for each of the first to pth IPs **200_1** to **200_p** may be independently set, and whether to allow the access of the user device **10** to each of the first to pth IPs **200_1** to **200_p** may also be independently determined.

[0074] FIG. 6 is a flowchart illustrating an example of an operating method of a security device.

[0075] In FIG. 6 and below, the configuration of the security device **100** will be described focusing on the implementation illustrated in FIG. 2; however, in the following descriptions, the configuration of the security device **100** may also be similarly applied to the implementation illustrated in FIG. 4.

[0076] Referring to FIG. 6, in operation **S610**, the security device **100** may receive an access request from the user device **10**. The access request may be a request of the user device **10** for allowing access for performing debugging on the plurality of IPs **200** included in the security device **100** and may include information required for authentication corresponding to the security mode set by the mode controller **110**.

[0077] In operation **S620**, the security device **100** may determine whether to allow the access, based on the security mode. The security device **100** may determine whether to allow the access of the user device **10** to the plurality of IPs **200**, through the authentication controller **120**, and a more detailed method thereof will be described below with reference to FIGS. 7 to 10.

[0078] In operation **S630**, the security device **100** may activate or deactivate the access of the user device **10** based on the determination result of operation **S620**. When it is determined in operation **S620** to allow the access of the user device **10** to the plurality of IPs **200**, the security device **100** may activate the access of the user device **10** through the access controller **130**. On the contrary, when it is determined in operation **S620** not to allow the access of the user device **10** to the plurality of IPs **200**, the security device **100** may deactivate the access of the user device **10** through the access controller **130**.

[0079] FIG. 7 is an example flowchart illustrating the operation of a user device and a security device when the security mode of the security device is a first security mode.

[0080] Referring to FIG. 7, when the security mode of the security device is the first security mode, the operation of the user device and the security device may be identified. The implementation of FIG. 7 will be described focusing on the main operations of the user device and the security device; however, incidental operations may be added thereto.

[0081] In operation S710, the security device **100** may set the security mode to the first security mode. The security device **100** may set the security mode based on the OTP value stored in the OTP memory **111** included in the mode controller **110**. In the implementation of FIG. 7, the OTP value stored in the OTP memory **111** may be “000”, and the mode controller **110** may set the security mode to the first security mode because the OTP value is “000”.

[0082] In operation S720, the user device **10** may identify the security mode of the security device **100**, and accordingly, it may be identified that the security mode of the security device **100** is the first security mode.

[0083] In operation S730, the user device **10** may transmit an access request to the security device **100**. In this case, because the security mode of the security device **100** is the first security mode, the user device **10** may transmit the access request without separate information required for authentication.

[0084] In operation S740, the security device **100** may activate the access of the user device **10**. In this case, because the security mode of the security device **100** is the first security mode, the authentication controller **120** may determine to allow the access without performing separate authentication, and accordingly, the access controller **130** may activate the access of the user device **10**.

[0085] FIG. 8 is an example flowchart illustrating the operation of a user device and a security device when the security mode of the security device is a second security mode.

[0086] Referring to FIG. 8, when the security mode of the security device is the second security mode, the operation of the user device and the security device may be identified. The implementation of FIG. 8 will be described focusing on the main operations of the user device and the security device; however, incidental operations may be added thereto.

[0087] In operation S810, the security device **100** may set the security mode to the second security mode. The security device **100** may set the security mode based on the OTP value stored in the OTP memory **111** included in the mode controller **110**. In the implementation of FIG. 8, the OTP value stored in the OTP memory **111** may be “001”, and the mode controller **110** may set the security mode to the second security mode because the OTP value is “001”.

[0088] In operation S820, the user device **10** may identify the security mode of the security device **100**, and accordingly, it may be identified that the security mode of the security device **100** is the second security mode.

[0089] In operation S830, the user device **10** may transmit a password to the security device **100**. In this case, because the security mode of the security device **100** is the second security mode, the access authority of the user device **10** may be authenticated through the password.

[0090] In operation S840, the security device **100** may verify the password. The security device **100** may perform a hash operation on the password through the encryption circuit **121** of the authentication controller **120**. The encryption circuit **121** may verify the password by determining whether the hash value of the password is equal to a prestored hash value.

[0091] In this case, when the hash value of the password obtained through the operation is equal to the prestored hash value, the encryption circuit **121** may determine to allow the access of the user device **10**. On the contrary, when the hash value of the password obtained through the operation is different from the prestored hash value, the encryption circuit **121** may determine not to allow the access of the user device **10**.

[0092] In operation S850, the security device **100** may activate or deactivate the access of the user

device **10** based on the verification result.

[0093] When it is determined in operation **S840** to allow the access of the user device **10**, the security device **100** may activate the access of the user device **10** through the access controller **130**. On the contrary, when it is determined in operation **S840** not to allow the access of the user device **10**, the security device **100** may deactivate the access of the user device **10** through the access controller **130**.

[0094] FIG. **9** is an example flowchart illustrating the operation of an authentication server, a user device, and a security device when the security mode of the security device is a third security mode.

[0095] Referring to FIG. **9**, when the security mode of the security device is the third security mode, the operation of the authentication server, the user device, and the security device may be identified. The implementation of FIG. **9** will be described focusing on the main operations of the authentication server, the user device, and the security device; however, incidental operations may be added thereto.

[0096] In operation **S910**, the security device **100** may set the security mode to the third security mode. The security device **100** may set the security mode based on the OTP value stored in the OTP memory **111** included in the mode controller **110**. In the implementation of FIG. **9**, the OTP value stored in the OTP memory **111** may be “011”, and the mode controller **110** may set the security mode to the third security mode because the OTP value is “011”.

[0097] In operation **S915**, the user device **10** may identify the security mode of the security device **100**, and accordingly, it may be identified that the security mode of the security device **100** is the third security mode.

[0098] In operation **S920**, the user device **10** may transmit a user ID to the authentication server **20**.

[0099] In operation **S925**, the authentication server **20** may identify the user authority based on the user ID. The user authority may be the authority for allowing the user corresponding to the user ID to access the security device **100**. In this case, when the authentication server **20** identifies that the user does not have the authority, subsequent operations may not be performed. When the authentication server **20** identifies that the user has the authority, operation **S930** and subsequent operations may be performed.

[0100] In operation **S930**, the authentication server **20** may transmit a public key to the user device **10**.

[0101] In operation **S935**, the user device **10** may transmit the public key received in operation **S930** to the security device **100**.

[0102] In operation **S940**, the security device **100** may verify the public key. The security device **100** may verify whether the public key is correct, through the encryption circuit **121**. In this case, when the encryption circuit **121** determines that the public key is not correct, subsequent operations may not be performed. On the contrary, when the encryption circuit **121** identifies that the public key is correct, operation **S945** and subsequent operations may be performed.

[0103] In operation **S945**, the security device **100** may transmit a public key approval message to the user device **10** as the public key is verified as being correct in operation **S940**.

[0104] In operation **S950**, the user device **10** may transmit the public key approval message received in operation **S945** to the authentication server **20**.

[0105] In operation **S955**, the authentication server **20** may generate a digital signature corresponding to the user ID upon receiving the public key approval message. For example, the authentication server **20** may generate a digital signature by using an Elliptic Curve Digital Signature Algorithm (ECDSA) method.

[0106] In operation **S960**, the authentication server **20** may transmit the digital signature generated in operation **S955** to the user device **10**.

[0107] In operation **S965**, the user device **10** may transmit the digital signature received in operation **S960** to the security device **100**.

[0108] In operation **S970**, the security device **100** may verify the digital signature. The security device **100** may verify whether the digital signature is correct through the encryption circuit **121**. [0109] When the digital signature is identified as being correct, the encryption circuit **121** may determine to allow the access of user device **10**. On the contrary, when the digital signature is identified as not being incorrect, the encryption circuit **121** may determine not to allow the access of the user device **10**.

[0110] In operation **S975**, the security device **100** may activate or deactivate the access of the user device **10** based on the verification result.

[0111] When it is determined in operation **S970** to allow the access of the user device **10**, the security device **100** may activate the access of the user device **10** through the access controller **130**. On the contrary, when it is determined in operation **S970** not to allow the access of the user device **10**, the security device **100** may deactivate the access of the user device **10** through the access controller **130**.

[0112] FIG. **10** is an example flowchart illustrating the operation of an authentication server, a user device, and a security device when the security mode of the security device is a fourth security mode.

[0113] Referring to FIG. **10**, when the security mode of the security device is the fourth security mode, the operation of the authentication server, the user device, and the security device may be identified. The implementation of FIG. **10** will be described focusing on the main operations of the authentication server, the user device, and the security device; however, incidental operations may be added thereto.

[0114] In operation **S1010**, the security device **100** may set the security mode to the fourth security mode. The security device **100** may set the security mode based on the OTP value stored in the OTP memory **111** included in the mode controller **110**. In the implementation of FIG. **10**, the OTP value stored in the OTP memory **111** may be “111”, and the mode controller **110** may set the security mode to the fourth security mode because the OTP value is “111”.

[0115] In operation **S1020**, the user device **10** may identify the security mode of the security device **100**, and accordingly, it may be identified that the security mode of the security device **100** is the fourth security mode.

[0116] In operation **S1030**, the user device **10** may transmit a user ID to the authentication server **20**.

[0117] In operation **S1040**, the authentication server **20** may identify the user authority based on the user ID. In this case, when the authentication server **20** identifies that the user does not have the authority, subsequent operations may not be performed. When the authentication server **20** identifies that the user has the authority, operation **S1050** and subsequent operations may be performed.

[0118] In operation **S1050**, the authentication server **20** may transmit a public key to the user device **10**.

[0119] In operation **S1060**, the user device **10** may transmit the public key received in operation **S1050** to the security device **100**.

[0120] In operation **S1070**, the security device **100** may verify the public key. The security device **100** may verify whether the public key is correct, through the encryption circuit **121**. In this case, when the encryption circuit **121** determines that the public key is not correct, subsequent operations may not be performed. On the contrary, when the encryption circuit **121** identifies that the public key is correct, operation **S1080** and subsequent operations may be performed.

[0121] In operation **S1080**, the security device **100** may generate a random value. The security device **100** may generate a random value through the random number generator **122** of the authentication controller **120**.

[0122] In operation **S1090**, the security device **100** may transmit the random value generated in operation **S1080** to the user device **10**.

[0123] In operation **S1100**, the user device **10** may transmit the random value received in operation **S1090** to the authentication server **20**.

[0124] In operation **S1110**, the authentication server **20** may generate a digital signature based on the random value. For example, the authentication server **20** may generate a digital signature by using an ECDSA method.

[0125] In operation **S1120**, the authentication server **20** may transmit the digital signature generated in operation **S1110** to the user device **10**.

[0126] In operation **S1130**, the user device **10** may transmit the digital signature received in operation **S1120** to the security device **100**.

[0127] In operation **S1140**, the security device **100** may verify the digital signature. The security device **100** may verify whether the digital signature is correct through the encryption circuit **121**.

[0128] When the digital signature is identified as being correct, the encryption circuit **121** may determine to allow the access of user device **10**. On the contrary, when the digital signature is identified as not being incorrect, the encryption circuit **121** may determine not to allow the access of the user device **10**.

[0129] In operation **S1150**, the security device **100** may activate or deactivate the access of the user device **10** based on the verification result.

[0130] When it is determined in operation **S1140** to allow the access of the user device **10**, the security device **100** may activate the access of the user device **10** through the access controller **130**. On the contrary, when it is determined in operation **S1140** not to allow the access of the user device **10**, the security device **100** may deactivate the access of the user device **10** through the access controller **130**.

[0131] While this specification contains many specific implementation details, these should not be construed as limitations on the scope of any invention or on the scope of what may be claimed, but rather as descriptions of features that may be specific to particular implementations of particular inventions. Certain features that are described in this specification in the context of separate implementations can also be implemented in combination in a single implementation. Conversely, various features that are described in the context of a single implementation can also be implemented in multiple implementations separately or in any suitable subcombination. Moreover, although features may be described above as acting in certain combinations, one or more features from a combination can in some cases be excised from the combination, and the combination may be directed to a subcombination or variation of a subcombination.

[0132] While the present disclosure has been shown and described with reference to implementations thereof, it will be understood that various changes in form and details may be made therein without departing from the spirit and scope of the following claims.

Claims

1. A security device configured to control access authority of a user device, the security device comprising: a mode controller configured to set, based on a one-time programmable (OTP) value, a security mode, the security mode indicating a method of authenticating the access authority of the user device; an authentication controller configured to determine, based on the security mode, whether to allow access of the user device; and an access controller configured to activate or deactivate the access of the user device based on a result of the determination of the authentication controller.
2. The security device of claim 1, wherein the OTP value includes one or more bits, and a value of each bit of the one or more bits is changeable in an increasing direction or a decreasing direction.
3. The security device of claim 2, wherein the mode controller is configured to change the security mode based on the OTP value being changed.
4. The security device of claim 3, wherein the OTP value includes $n-1$ bits, n being a natural

number greater than or equal to 2, and wherein the mode controller is configured to set the security mode to one of first to n .sup.th security modes based on a number of bits with a first logic value among the $n-1$ bits included in the OTP value.

5. The security device of claim 4, wherein the number of bits with the first logic value among the $n-1$ bits included in the OTP value is k , k being a natural number of 0 to $n-1$, wherein the mode controller is configured to set the security mode to a $(k+1)$.sup.th security mode, and wherein an i .sup.th security mode has higher security strength than a j .sup.th security mode, i being a natural number of 2 to n , j being a natural number of 1 to $n-1$, and j being less than i .

6. The security device of claim 1, wherein the mode controller is configured to set the security mode to one of first to fourth security modes based on the OTP value, wherein the first security mode is a mode of allowing the access of the user device without authentication, wherein the second security mode is a mode of authenticating the access authority of the user device by using a password, wherein the third security mode is a mode of authenticating the access authority of the user device by using a digital signature, and wherein the fourth security mode is a mode of authenticating the access authority of the user device by using the digital signature and a random value.

7. The security device of claim 6, wherein the authentication controller is configured to based on the security mode being the second security mode, determine whether to allow the access of the user device based on the password, based on the security mode being the third security mode, determine whether to allow the access of the user device based on the digital signature, and based on the security mode being the fourth security mode, determine whether to allow the access of the user device based on the digital signature and the random value.

8. A security device configured to control access authority of a user device to a plurality of intellectual properties (IPs), the security device comprising: a mode controller configured to set a security mode of the user device for each IP of the plurality of IPs based on each one-time programmable (OTP) value of a plurality of OTP values; an authentication controller configured to determine, based on the security mode, whether to allow access of the user device to each IP of the plurality of IPs; and an access controller configured to activate or deactivate the access of the user device to each IP of the plurality of IPs based on a result of the determination of the authentication controller.

9. The security device of claim 8, wherein each OTP value of the plurality of OTP values includes one or more bits, and a value of each bit of the one or more bits is changeable in an increasing direction or a decreasing direction.

10. The security device of claim 8, wherein a number of the plurality of OTP values equals to a number of the plurality of IPs, and wherein based on an m .sup.th OTP value being changed, the mode controller is configured to change the security mode for an m .sup.th IP corresponding to the m .sup.th OTP value, m being a natural number.

11. The security device of claim 10, wherein each OTP value of the plurality of OTP values includes $n-1$ bits, n being a natural number greater than or equal to 2, and wherein the mode controller is configured to set the security mode for the m .sup.th IP to one of first to n .sup.th security modes based on a number of bits with a first logic value among the $n-1$ bits included in the m .sup.th OTP value.

12. The security device of claim 11, wherein the number of bits with the first logic value among the $n-1$ bits included in the m th OTP value is k , k being a natural number of 0 to $n-1$, wherein the mode controller is configured to set the security mode to a $(k+1)$.sup.th security mode, and wherein an i .sup.th security mode has higher security strength than a j .sup.th security mode, i being a natural number of 2 to n , j being a natural number of 1 to $n-1$, and j being less than i .

13. The security device of claim 8, wherein the mode controller is configured to set the security mode for an m .sup.th IP to one of first to fourth security modes based on an m .sup.th OTP value, m being a natural number, wherein the first security mode is a mode of allowing the access of the user

device to each IP of the plurality of IPs without authentication, wherein the second security mode is a mode of authenticating the access authority of the user device to each IP of the plurality of IPs by using a password, wherein the third security mode is a mode of authenticating the access authority of the user device to each IP of the plurality of IPs by using a digital signature, and wherein the fourth security mode is a mode of authenticating the access authority of the user device to each IP of the plurality of IPs by using the digital signature and a random value.

14. The security device of claim 13, wherein the authentication controller is configured to based on the security mode being the second security mode, determine whether to allow the access of the user device to each IP of the plurality of IPs based on the password, based on the security mode being the third security mode, determine whether to allow the access of the user device to each IP of the plurality of IPs based on the digital signature, and based on the security mode being the fourth security mode, determine whether to allow the access of the user device to each IP of the plurality of IPs based on the digital signature and the random value.

15. The security device of claim 8, wherein the authentication controller is configured to sequentially determine whether to allow the access of the user device to each IP of the plurality of IPs based on the security mode of each IP of the plurality of IPs.

16. The security device of claim 8, wherein the access controller is configured to independently activate or deactivate the access of the user device to each IP of the plurality of IPs.

17. A system-on-chip (SoC) comprising: a plurality of intellectual properties (IPs); and a security device configured to control access authority of a user device to the plurality of IPs, wherein the security device comprises: a mode controller configured to set, based on a one-time programmable (OTP) value, a security mode, the security mode indicating a method of authenticating the access authority of the user device to the plurality of IPs; an authentication controller configured to determine, based on the security mode, whether to allow access of the user device to the plurality of IPs; and an access controller configured to activate or deactivate the access of the user device to the plurality of IPs based on a result of the determination of the authentication controller.

18. The SoC of claim 17, wherein the OTP value includes one or more bits, and a value of each bit of the one or more bits is changeable in an increasing direction or a decreasing direction.

19. The SoC of claim 18, wherein the mode controller is configured to change the security mode based on the OTP value being changed.

20. The SoC of claim 17, wherein the mode controller is configured to set the security mode to one of first to fourth security modes based on the OTP value, wherein the first security mode is a mode of allowing the access of the user device to the plurality of IPs without authentication, wherein the second security mode is a mode of authenticating the access authority of the user device to the plurality of IPs by using a password, wherein the third security mode is a mode of authenticating the access authority of the user device to the plurality of IPs by using a digital signature, and wherein the fourth security mode is a mode of authenticating the access authority of the user device to the plurality of IPs by using the digital signature and a random value.
